Topological Analysis of Ocean Sound Speed Profiles

Matthew LeDuc, Arnaldo Pereira, Samuel Zhang

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1 Introduction

While energy can be transmitted through the air via electromagnetic waves without much propagation loss, the ocean is nearly opaque to electromagnetic radiation. However, the ocean is much more transparent to sound than the atmosphere is. While most electromagnetic radiation gets absorbed within 100m of the source, acoustic waves can and do travel hundreds of kilometers [7]. Because of this, most ocean remote sensing applications make use of acoustic rather than electromagnetic waves. Because of its widespread use, understanding acoustic propagation in the ocean is of the utmost importance for a wide variety of applications. In this paper, we intend to explore the applications of the techniques of topological data analysis to underwater sound speed profiles. We investigate the persistent homologies in this data in an attempt to uncover further structure in the space of ocean sound speed profiles. While there have been several investigations into classification of oceanographic features using conventional clustering algorithms (such as [12] and [14]) we are unaware of any investigations into these datasets using topological techniques.

2 Underwater Acoustics

The propagation of sound through the ocean is dictated by the sound speed profile. Sound speed in water is a complicated relationship of the depth, temperature, and salinity. However, the temperature and salinity vary over both large and small spatial and temporal scales, which can make classification difficult [14].

There are several different equations that can be used to convert temperature, salinity, and depth into sound speed, but one of the most commonly used is the Chen-Millero-Li sound speed equation presented in [2] and [15]. Near the surface the change in sound speed is determined by the temperature gradient. Sound speed decreases with temperature, so as the water temperature decreases so does the sound speed. However, deeper into the ocean, the temperature gradient eventually flattens out and sound speed begins to increase again as the pressure becomes the dominant term in the sound speed equation. This generally happens around 700m depth, but there is significant variation from this [7]. In the polar regions where water temperature is more or less constant, sound speed will be lowest at the surface and increase with depth. However, near the Equator or other regions of warm water, the temperature will not be constant, so sound speed may not reach a minimum until much deeper [21]. There is also some seasonal dependence, especially at mid-latitudes where temperatures can vary significantly between seasons.

The depth at which sound speed is slowest, known as the sound channel axis depth, is one of if not the most important features of a sound speed profile. As acoustic wavefronts travel through the ocean and come to an interface between two layers of varying sound speeds, they will refract in accordance with Snell's Law

$$\frac{\cos(\theta_1)}{c_1} = \frac{\cos(\theta_2)}{c_2} \tag{1}$$

where c_i is the sound speed in layer i and θ_i is the grazing angle at the interface in layer i. This means that wavefronts will always tend to curve toward the layer with lower sound velocity. From this, we can find a critical angle θ_c such that wavefronts incident on the interface at θ_c cannot pass into the next layer:

$$\theta_c = \arccos \frac{c_2}{c_1} \tag{2}$$

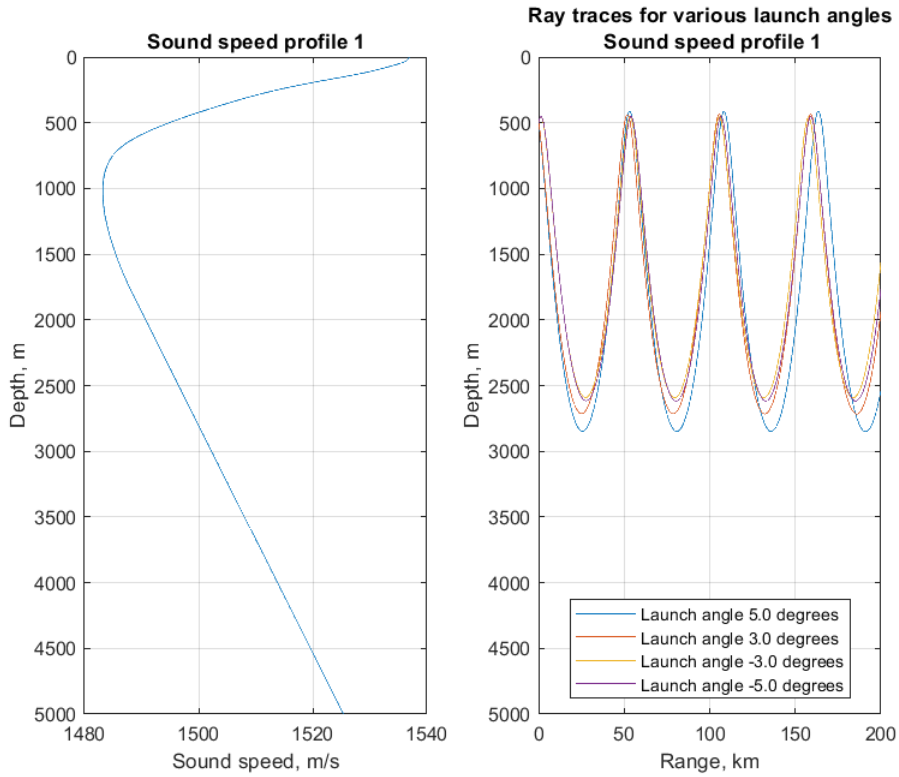


Figure 1: An example sound speed profile admitting entirely waterborne propagation for certain launch angles

This leads to the behavior seen in figure 1, where wavefronts emitted at small angles from a source at 500m depth never reach the surface or the ocean floor. Because of this, these rays will not experience scattering losses from reflection off the surface or ocean floor, and will therefore propagate much further than they would otherwise. It is believed that some species of whales leverage this for communication over long distances [7].

Comparing this to figure 2 we see a meaningful difference: wavefronts are not strongly refracted, so they reflect off the bottom and surface of the ocean. This results in significant loss in energy due to scattering, meaning these wavefronts will not travel as far as the ones in figure 1.

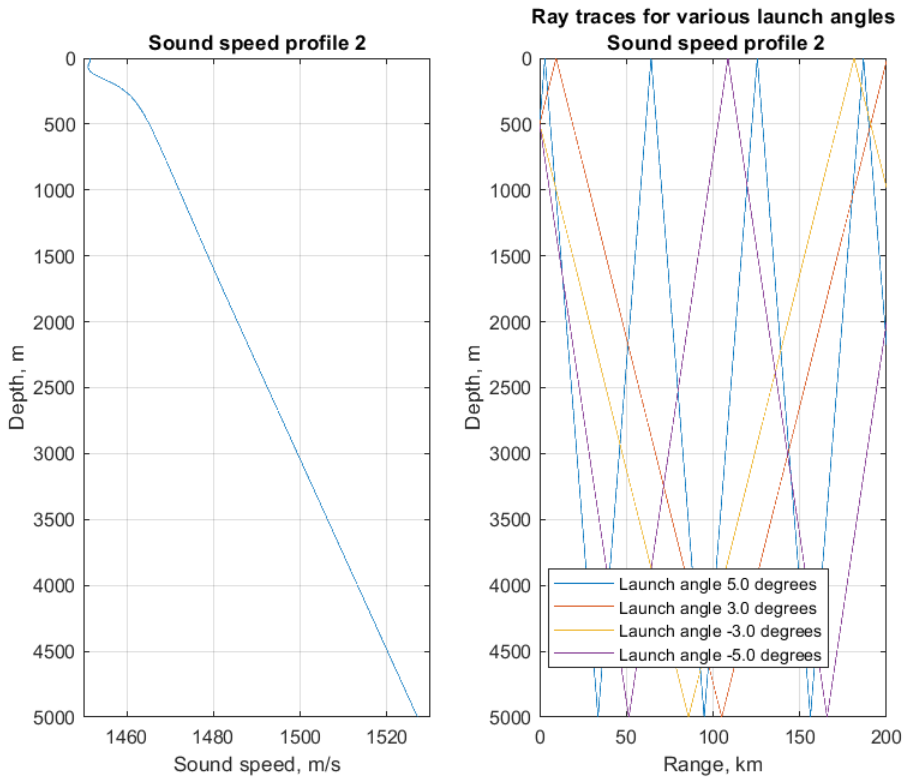


Figure 2: A sound speed profile that admits no waterborne propagation

3 Data

Our data comes from NASA ECCO (Estimating the Circulation & Climate of the Ocean) databases [4]. This database contains both the outputs and inputs to the NASA ECCO model. We will be using the model input data for our analysis. This data comes from several thousand ARGO profiling floats, which are small buoyancy-driven devices that travel through the ocean collecting and reporting data on local temperature, salinity, and pressure at 54 different depths up to 2000m depth. At present, there are roughly 12,000 temperature and salinity profiles recorded every month by these floats [1]. By using the Chen-Millero-Li sound speed equation, we can use these temperature, salinity, and pressure profiles to generate sound speed profiles.

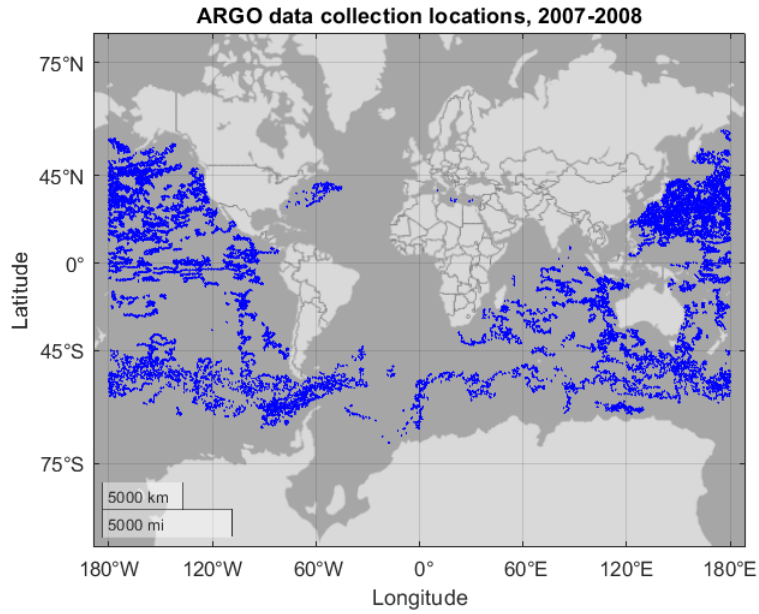


Figure 3: Location of ARGO profiles taken from 2007-2008

4 Implementation

Below is a brief summary of the computational approach taken:

1. In each longitude/latitude "box", we generate an empirical cumulative distribution function of sound channel axis depth; each such C.D.F. is a point in the point cloud.

2. Between each pair of points, compute the pairwise distance (detailed in Section 5 - Analysis), and sort all edges (1-simplices) by increasing distance.

3. Add all points (0-simplices) to the initial subcomplex at diameter 0. These are the leaves in a "face-coface" tree data structure, where each simplex has all of its faces as children.

4. Generate a filtration by iterating over a sample of diameters and adding edges by increasing diameter. Using the tree data structure, for each simplex we keep track how many of its faces have been added so far; once all the faces have been included, we add the coface as it emerges during this process of the Vietoris-Rips construction.

5. As described by Edelsbrunner and Harer [6], form an M×M matrix, where M is the number of simplices in the final (full) complex. Each column corresponds to a simplex, and the rows correspond to the faces. Perform column reduction on the matrix, and record which columns were added to which other ones; the zero columns represent cycles of the columns which were added together. Each pivot at entry (r,c) in the matrix indicates that the simplex corresponding to column c killed the homology class that was born by the simplex from column r. As an optimization, we use a sparse matrix representation, which is naturally efficient using set intersection operations in $\mathbb{Z}_2$, and reduces the overall space complexity.

6. For each cycle, determine whether it is bounded, i.e. colloquially "filled in" by checking whether it is linearly dependent with respect to the image space of the respective (linear) boundary map. More precisely, for a cycle in the kernel (Z_k) of the kth boundary operator, create a row vector where the ith entry is 1 if the ith simplex is in that chain (and 0 otherwise), then append the vector to the bottom of the matrix for the image space (for B_k), and if the rank increases then that new vector is linearly independent of the original row space. These cycles are different from the identity coset [0] in the quotient group Z_k/B_k.

7. For each cycle that is not in the boundary (as found in step 6), compute the persistence by subtracting the death diameter from the birth diameter (in other sources, "time") obtained during step 5. Over its lifetime, each persistent homology can be located across multiple abstract simplicial complexes in the filtration which we obtained after step 4.

5 Analysis

Our processing of the data will be focused around the calculation of cumulative density functions for sound channel axis depth. Each empirical CDF is generated from a subset of data in a "bounding box" (containing all points between two latitude and longitude values), thus partitioning Earth's surface. Once we generated the distributions, the following distance functions were used to generate distance matrices and perform filtrations.

5.1 Kolmogorov-Smirnov Test

The first metric we will analyze our data with comes from the two-sample Kolmogorov-Smirnov Test. This test is commonly used to establish whether or not two empirical cumulative density functions came from the same underlying probability distribution [13]. Given two cumulative density functions F_1 and F_2 in the space of functions $F : X \to [0, 1]$ following the usual conditions for probability distributions on X (in our case, restricted to $X \subset \mathbb{R}$), the Kolmogorov-Smirnov test statistic D is given by

$$D = \sup_{x \in X} |F_1(x) - F_2(x)| \tag{3}$$

In the case where F_1 and F_2 are empirical cumulative distribution functions consisting of n data points the Kolmogorov-Smirnov test statistic is equivalent to the L_∞ metric on $[0, 1]^n$.

One useful property of this test is that it comes with a test statistic. Given two empirical CDFs F_1 and F_2 made with n datapoints, the CDFs are considered different at the α significance level if

$$D_\alpha \geq \sqrt{-\frac{1}{n} \log \frac{\alpha}{2}} \tag{4}$$

Here and throughout this paper, log refers to the natural logarithm. For $n = 54$ datapoints, this gives $D_{0.05} \approx 0.26$.

5.2 Kullback-Leibler Divergence

In addition to the Kolmogorov-Smirnov test, which is a metric on the space of cumulative distribution functions of the form $F : \mathbb{R} \to [0, 1]$, we also compared the data using the Kullback-Leibler divergence [10] on the empirical probability mass functions for our observed data. As a reminder, a probability mass function $f : X \to [0, 1]$ for a discrete probability distribution has the following properties:

$$f(x) \geq 0 \ \forall x \in X \tag{5a}$$

$$\sum_{x \in X} f(x) = 1 \tag{5b}$$

and where $X \subset \mathbb{R}$, as in our data distributions, the cumulative distribution function F, which is used in the Kolmogorov-Smirnov test, is related to the probability mass function by

$$F(x) = \sum_{z \leq x} f(z) \tag{6}$$

The Kullback-Leibler divergence for two probability mass functions $f_1 and f_2$ on a X is given by

$$D_{KL}(f_1, f_2) = \sum_{x \in X} f_1(x) \log \frac{f_1(x)}{f_2(x)} \tag{7}$$

However, while (7) satisfies $D_{KL}(f_i(x), f_i(x)) = 0$, it is not symmetric and it does not satisfy the triangle inequality and therefore is not a metric [9]. However, we can define the symmetric Kullback-Leibler Divergence as [10]

$$D_{SKL}(f_1, f_2) = \frac{1}{2} \left(D_{KL}(f_1, f_2) + D_{KL}(f_2, f_1) \right) = \frac{1}{2} \sum_{x \in X} \left(f_1(x) \log \frac{f_1(x)}{f_2(x)} + f_2(x) \log \frac{f_2(x)}{f_1(x)} \right) \tag{8}$$

While this still does not satisfy the triangle inequality, it is now symmetric. We claim that this expression is non-negative and equal to zero only when $f_1 = f_2$. To prove this, we present a canonical proof using Jensen's Inequality.

Jensen's Inequality states that, for a convex function ϕ and scalar weights α_i,

$$\phi\left(\frac{\sum_i \alpha_i x_i}{\sum_i \alpha_i}\right) \leq \frac{\sum_i \alpha_i \phi(x_i)}{\sum_i \alpha_i} \tag{9}$$

[8]. We use Jensen's inequality to prove that the KL Divergence applied to probability mass functions f_1 and f_2 is non-negative. We prove this fact for the restriction of the KL Divergence to a subset X of the domains of f_1 and f_2 on which both functions have nonzero total probability mass; that is, $f_1(x)$ and $f_2(x)$ are non-zero for at least some (possibly different for each function) $x_1, x_2 \in X$. First, we re-write equation (7) as

$$D_{KL}(f_1, f_2) = -\sum_{x \in X} f_1(x) \log \frac{f_2(x)}{f_1(x)} \tag{10}$$

Since the logarithm is a convex function, we can apply Jensen's Inequality to the equation above to show that

$$\sum_{x \in X} -f_1(x) \log \frac{f_2(x)}{f_1(x)} \geq -\log\left(\sum_{x \in X} f_1(x) \frac{f_2(x)}{f_1(x)}\right) = -\log\left(\sum_{x \in X} f_2(x)\right) \geq 0 \tag{11}$$

The last inequality follows from the fact that $\sum_{x \in X} f_2(x)$ is a sum taken over a subset with nonzero total probability mass for both f_1 and f_2, and therefore $0 < \sum_{x \in X} f_2(x) \leq 1$, which means its logarithm is negative. It follows that $D_{KL}(f_1, f_2) \geq 0$.

We show now that if X is the union of the domains of f_1 and f_2, then $D_{KL} = 0$ iff $f_1(x) = f_2(x)$ for all $x \in X$. Notice that $f_1(x) \geq 0$ and $\frac{f_2(x)}{f_1(x)} \geq 0$ for all x, since f_1 and f_2 are probability mass functions. And because the sum over the image of a probability mass function must equal 1, $f_1(x)$ and $f_2(x)$ must each be strictly greater than 0 for at least some $x \in X$ (not necessarily the same x for both). Therefore, $\sum_{x \in X} -f_1(x) \log \frac{f_2(x)}{f_1(x)} = 0$ iff $\log \frac{f_2(x)}{f_1(x)} = 0$ for all x, *i.e.*, iff $f_1 = f_2$.

To show that the symmetric Kullback-Leibler Divergence does not satisfy the triangle inequality, we consider three probability distributions on the space $X = \{1, 2, 3\}$ given by

$$\begin{aligned} f(x) &= \left[\frac{1}{2}, \frac{1}{2} - 10^{-10}, 10^{-10}\right] \\ g(x) &= \left[\frac{1}{6}, \frac{1}{6}, \frac{2}{3}\right] \\ h(x) &= \left[\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right] \end{aligned} \tag{12}$$

Then $D_{SKL}(f, g) \approx 11.857 > D_{SKL}(f, h) + D_{SKL}(g, h) \approx 4.7747 + 6.2019 = 10.9766$. Since $D_{SKL}(f, g) > D_{SKL}(f, h) + D_{SKL}(g, h)$, D_{SKL} does not satisfy the triangle inequality.

Since the symmetric KL divergence satisfies symmetry and $D_{SKL} \geq 0$ with equality only when $f_1(x) = f_2(x)$ but does not obey the triangle inequality, it is a semimetric [20]. Due to an error the implementation used to generate previous results does not satisfy the identity of discernibles, therefore our implementation of this is a premetric [16]. We will discuss this in more detail in the results section.

5.3 Jensen-Shannon Distance

The Jensen-Shannon distance is closely related to the Kullback-Leibler divergence. To derive this measure, we start from the Kullback-Leibler divergence in equation (7). First, given probability distributions f_1 and f_2 we define $m(x) = \frac{1}{2}(f_1(x) + f_2(x))$, which is their midpoint. Note that $m(x) \neq 0$ unless both $f_1(x)$ and $f_2(x) = 0$, since both are non-negative functions. Using this, we define the Jensen-Shannon divergence as

$$d_{JS}(f_1, f_2) = \frac{1}{2} \sum_{x \in X} f_1(x) \log \frac{f_1(x)}{m(x)} + f_2(x) \log \frac{f_2(x)}{m(x)} \tag{13}$$

We have shown that the Kullback-Leibler Divergence is non-negative and equal to zero only when $f_1 = f_2$, and the same proof holds for the Jensen-Shannon Divergence as well. In addition it is easily verified that the Jensen-Shannon Divergence is symmetric. But the Jensen-Shannon Divergence has another interesting property: The Jensen-Shannon Distance given by

$$D_{JS}(f_1, f_2) = \sqrt{d_{JS}(f_1, f_2)} \tag{14}$$

is a metric. [3]. The proof contained in [3] that the Jensen-Shannon distance satisfies the triangle inequality is in the appendix. It also shares a nice property with the Kolmogorov-Smirnov Test: for all probability distributions f_1 and f_2, $0 \leq D_{JS}(f_1, f_2) \leq \sqrt{\log(2)}$.

6 Results

In order to investigate structure in the distributions of sound channel axis depths, we divided 2000 randomly selected ARGO datapoints from 2007-2010 into 3 deg latitude by 3 deg longitude boxes, and generated a cumulative density function for each box with sufficient sample support. This resulted in 45 different cumulative density functions, with the location of each populated box marked by an 'x' in figure 4. By comparison with figure 3 we can see that the distribution of samples follows the distribution of the ARGO data reasonably well.

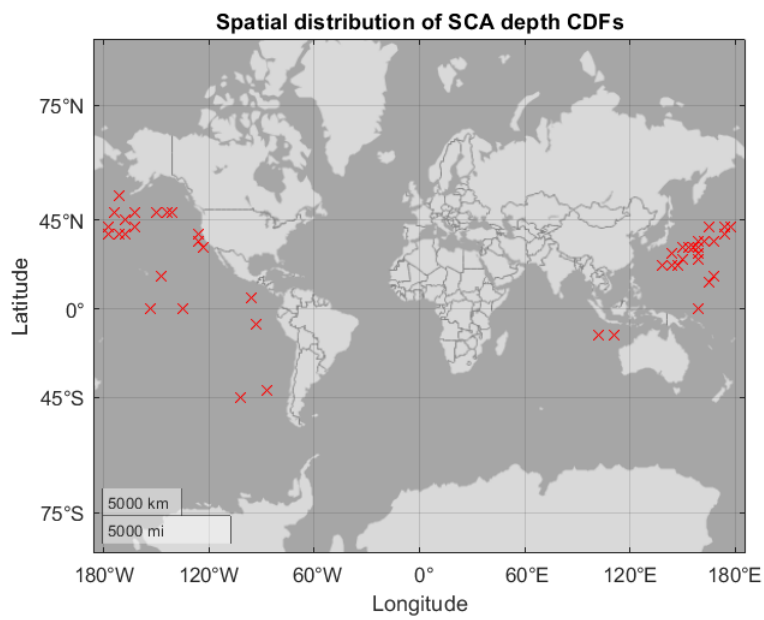


Figure 4: Spatial distribution of SCA depth CDFs

By using the Kolmogorov-Smirnov test as our metric and building a Vietoris-Rips complex by performing filtrations, we can examine the topology of the distribution by calculating homologies. Some pre-processing in Ripser [22] suggested that there was little to no structure beyond H_1 regardless of the metric we used so we focused our computations on simplices with dimension less than 3. Our attention is drawn to the homologies at filtration diameter $d = 0.32$. At $d = 0.32$ we see the single most persistent hole through all the filtrations. We will focus on this subcomplex to start.

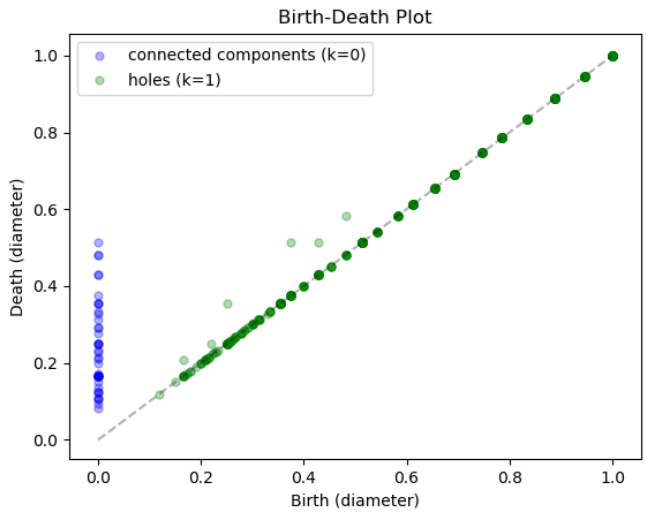


Figure 5: The persistence diagram using the KS test of H_0 and H_1 for a random draw of data over the world.

Since our data lives in $\mathbb{R}^{54}$, we plot each data point by mapping the distribution to its mean and variance. By using this mapping, we generate the plot in figure 6. This subcomplex has four connected components containing more than one point and one hole. As we hoped, CDFs with similar means and variances are grouped together. By plotting these large connected components on the globe, we can see the geographic locations of the datapoints that were grouped together (Figure 7).

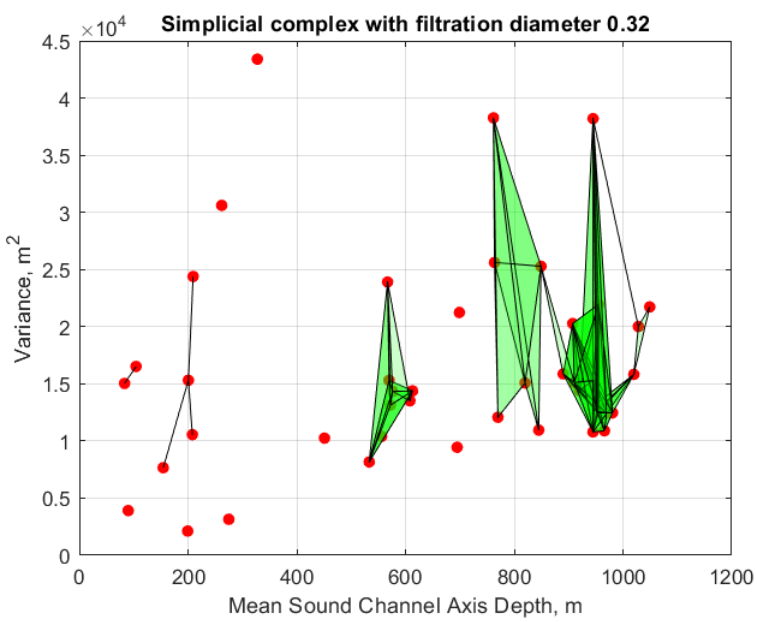


Figure 6: Vietoris-Rips complex with filtration diameter 0.32 for a random draw of data from across the whole world.

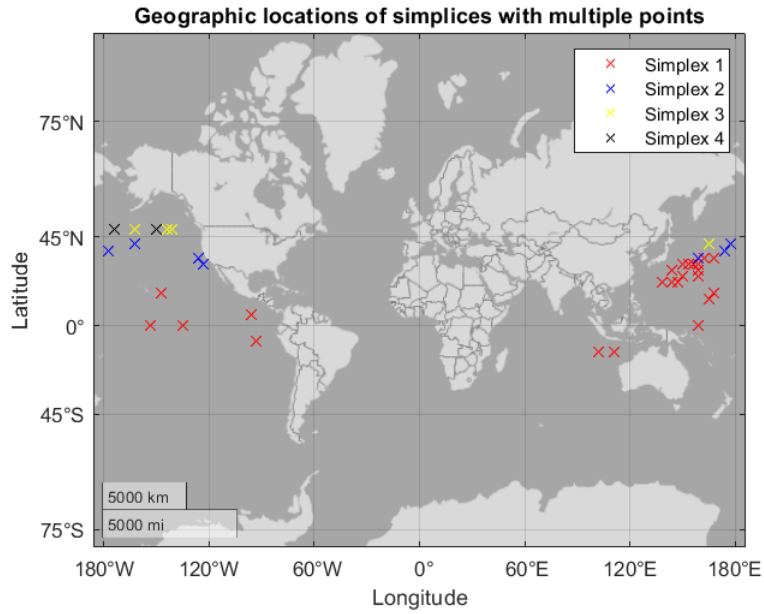


Figure 7: Color-coded connected components placed on the globe

For the most part, these results are expected. Profiles from latitudes between 20 deg N and 30 deg N are grouped together, with three other smaller groups of profiles whose sound channel axes are not as deep or invariant as the others. On closer inspection of the Northern Pacific near Japan, we see a potentially interesting result. There is a significant amount of variation in that area that respects what appears to be a boundary around 35N by 160E, the approximate location of the intersection of the Oyashio and Kuroshio currents [17].

In addition, we looked at the Cyclostationary Empirical Orthogonal Function (CSEOF) Reconstructed Sea Level data generated by NASA JPL and the University of Colorado [5]. This data set contains data on sea surface height anomaly (SSHA), or the difference between the sea level in a region and the mean sea level. SSHA is correlated with water temperature [7]. This data shows that the two large connected components are mostly in regions where sea surface height is different, so there are differences in water temperature.

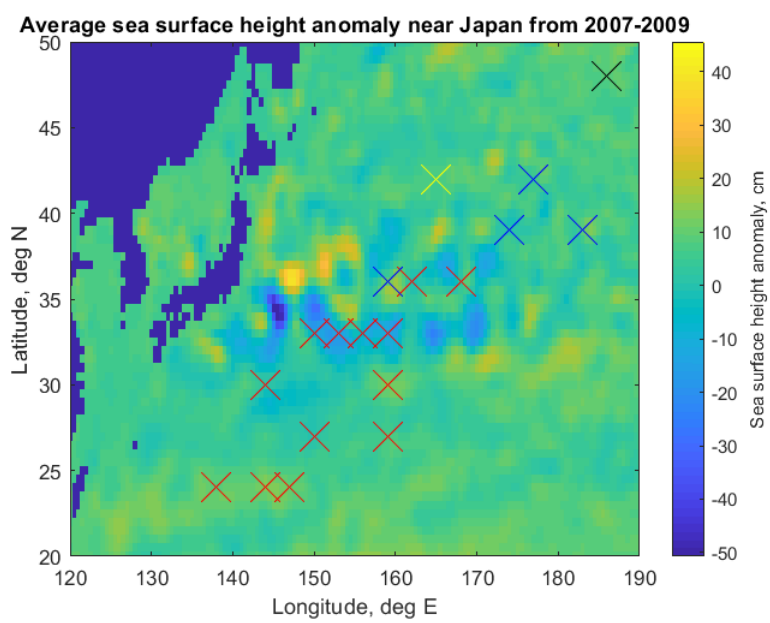


Figure 8: Mean sea surface height anomaly near Japan, 2007-2009

The dividing line between the two clusters is at approximately the latitude line where the currents meet (see figure (9)). To investigate this further, we will narrow our focus to the area bounded by 50Deg N, 30 deg S, 135 deg E, and 170 deg E and the cumulative density functions for each 1deg by 1deg lat/lon box contained therein.

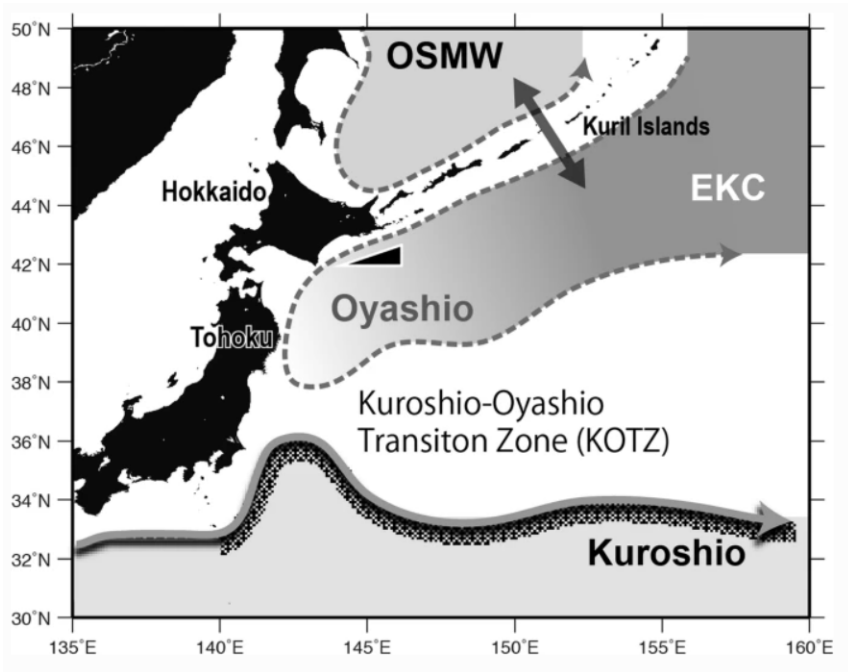
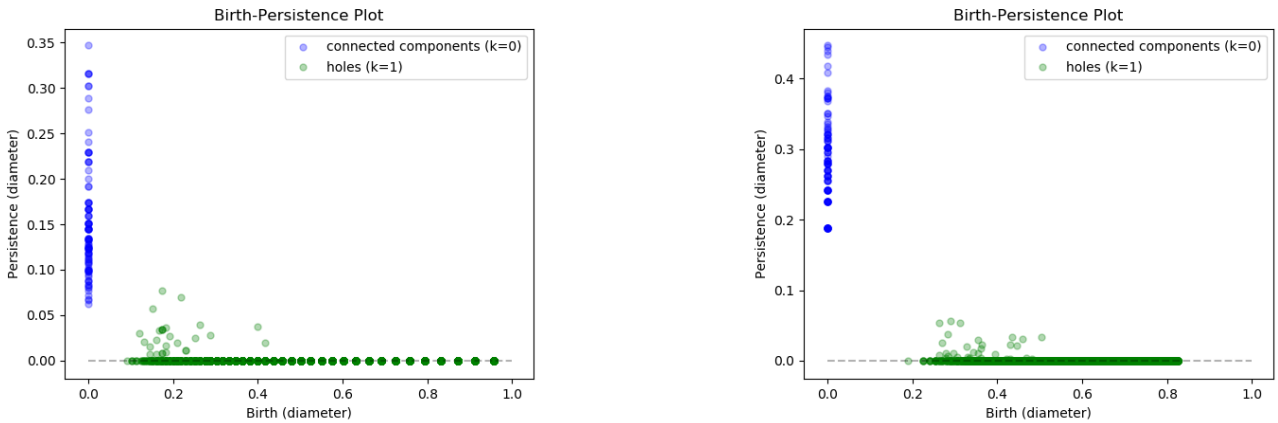


Figure 9: The approximate paths of the Kuroshio and Oyashio currents and their boundary, reprinted from [17]

6.1 Kuroshio and Oyashio currents

There are several reasons why the dynamics of the ocean near the intersection of the Kuroshio and Oyashio currents might be of interest. The region is known for its abundance of fish, which tend to spawn in the warm waters of the Kuroshio current and then swim to the transitional region to take advantage of the nutrient rich waters of the Oyashio current. As the global climate continues to change this transition region is expected to move northward, which could have a significant effect on the fishing industry in Japan [17].

By binning ARGO sound speed profiles into 1deg by 1deg lat/lon boxes and only forming a cumulative density function where there were more than 4 sound speed profile measurements, we were able to divide the data into approximately 300 distributions. We chose a random sample of 100 of these distributions to analyze. Doing this, we generate the two persistence diagrams shown in figure 11 using the Kolmogorov-Smirnov and Jensen-Shannon metrics.



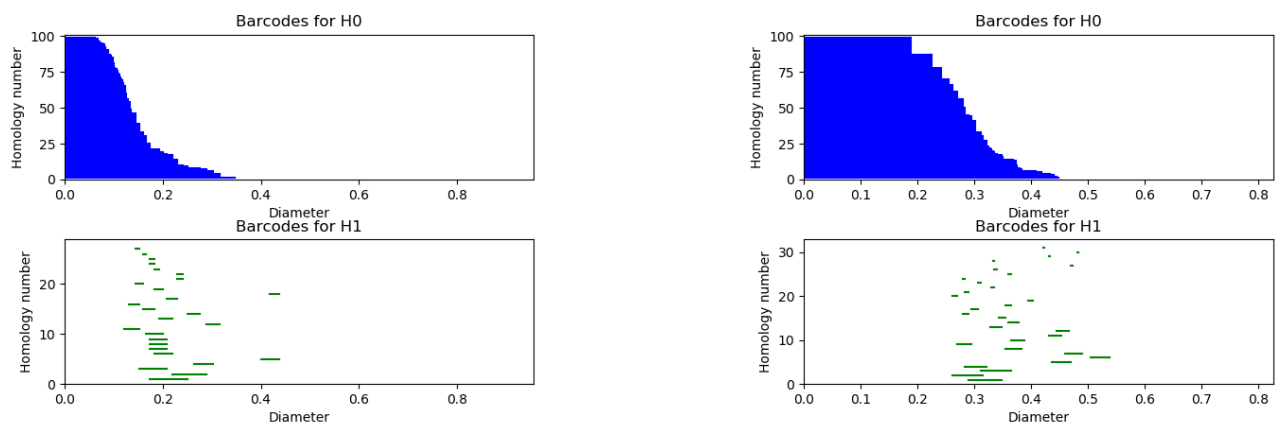
(a) Kolomogorov-Smirnov distance

(b) Jensen-Shannon distance

Figure 10: Birth-Persistence diagrams for the Kolmogorov-Smirnov and Jensen-Shannon metrics

These diagrams show that most holes are born and die out fairly early. There are some holes that last notably longer than others however, and we will investigate those. In particular, we will investigate the homologies near $d = 0.2$ generated by the Kolmogorov-Smirnov metric and $d = 0.3$ using the Jensen-Shannon metric.

In addition to the birth-persistence plots generated in figure 11, we can examine barcode plots to give more insight into the structure of our dataset. The first things we notice are the aforementioned holes near $d = 0.2$ and $d = 0.3$. However, there is not much underlying structure beyond this as most 1-homologies have persistence < 0.1. Regardless, we will examine the complexes containing these holes to see if there is deeper meaning to be pursued.



(a) Kolomogorov-Smirnov distance

(b) Jensen-Shannon distance

Figure 11: Barcode diagrams for the Kolmogorov-Smirnov and Jensen-Shannon metrics

Our first investigation is into the homologies that appear around a filtration diameter $d = 0.2$ using the Kolmogorov-Smirnov metric. Looking at the filtration with diameter $d = 0.21$ we see immediately a large hole where distributions with a mean of approximately $650m$ and a variance of approximately $4 \times 10^4 m^2$ would be (figure 12). It in unclear what the significance of this hole might be. The fact that its persistence is small compared to that of the larger connected components ($\approx 0.1 vs \approx 0.3$) suggests that it may be an artifact of the spatial resolution of our data. However, there is another possibility: our data is drawn from an area where a warm and cold current intersect, and therefore the dynamics of these areas are likely to be more robust to seasonal changes and therefore the distributions of sound channel axis depth might have a lower variance. This is supported by the fact that most distributions have a lower variance (by a factor of 2 or more!) than the distributions that would be in the hole would have. In addition, a mean depth of $650m$ is near the middle of the distribution of mean sample depths, and we might expect a region that was warm sometimes and cold other times to have a mean sound channel axis depth

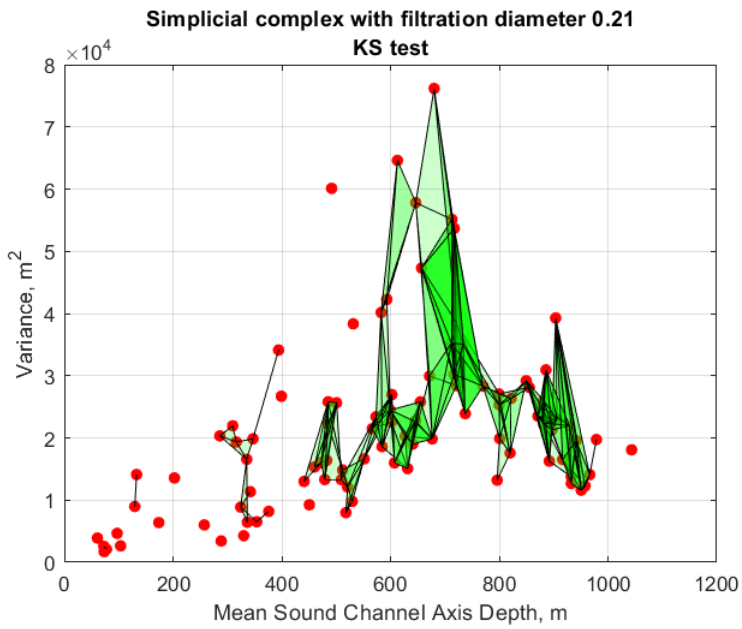
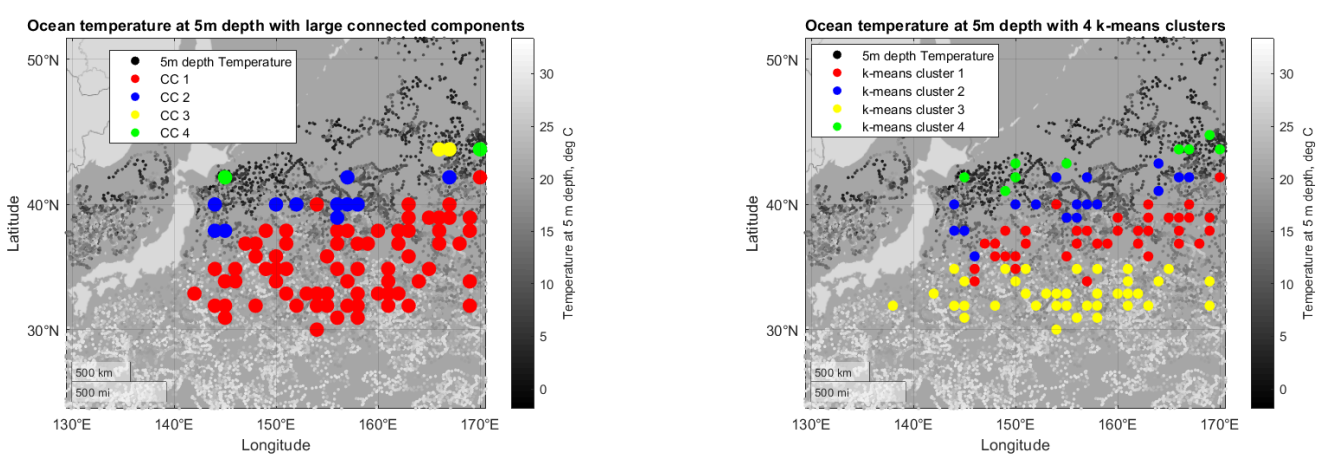


Figure 12: The subcomplex with diameter $d = 0.21$ generated using the Kolmogorov-Smirnov Test

that was somewhat in the middle of the two warm and cold water extremes. However, this is speculation and more investigation would be needed to determine whether this is a real feature of the environment or an artifact of spatial resolution.

Looking at this subcomplex, we can see that there are four large connected components (i.e. connected components containing more than one point). These components are shown in figure 13. Notice that the boundary between the largest components (colored red and blue) appears to respect the boundary between the transition zone and the Oyashio current around 40 degrees N. This can be seen via the greyscale colormap on the image, where darker colors represent cooler water and lighter colors represent warmer waters. The split between the components happens in an area where the water temperature changes quickly from approximately 15-20 deg Celsius to 5-10 deg Celsius. This structure is also visible when applying k-means with 4 clusters. However, k-means using the Euclidean metric also separates the distributions near the Kuroshio current from the ones near the transition zone between the two currents, while this subcomplex does not.



(a) Large connected components with water temperature at 5m depth

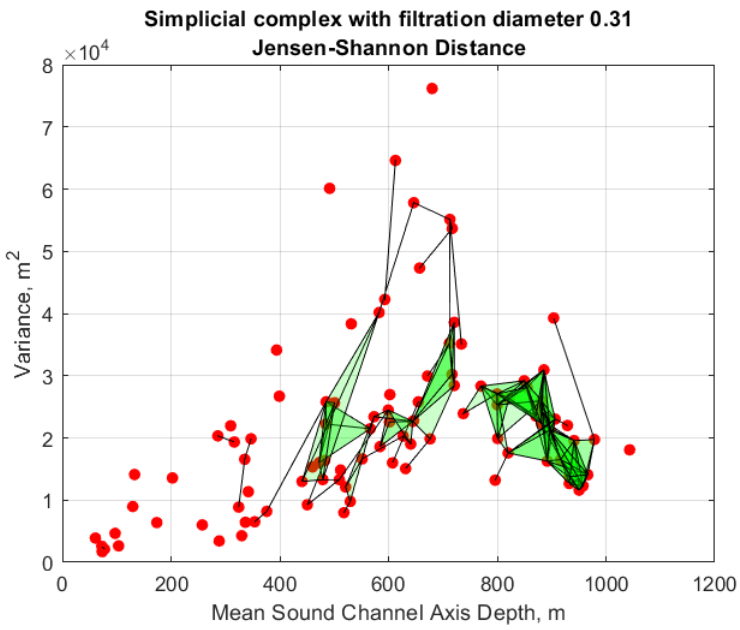
(b) Result of a k-means analysis with 4 clusters

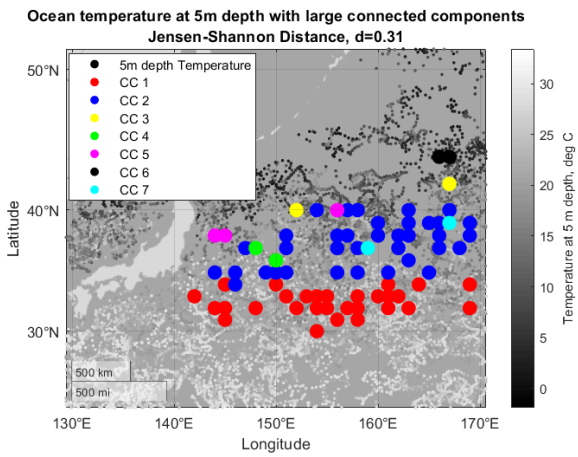
Figure 13: Clusters generated via a filtration using the KS metric and via k-means

We now turn our attention to the results derived by doing filtrations on the Jensen-Shannon distance. Based on the barcodes, it seems the most interesting structure is at $d = 0.3$ where there are 4 holes of approximately the same persistence and $d = 0.4$, where we see the most persistent 0-homology other than the set of all points.

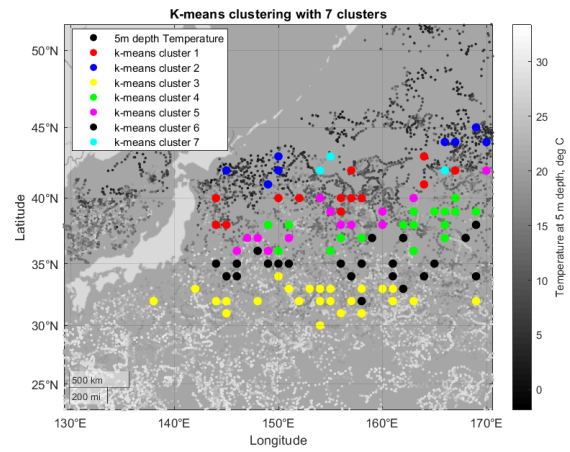
Upon examining the filtrations near $d = 0.3$ we notice some interesting behavior: One of the holes in the dataset is located in the same place as it was in the filtrations done using the Kolmogorov-Smirnov metric, and in addition has similar persistence. This filtration is show in figure **??**. However, unlike the filtration generated using the Kolmogorov-Smirnov metric, this filtration finds seven large connected components that exist at the same time as the hole instead of four. To visualize what this means in the real world, we plot the large connected components on the globe as before.

First, we notice that the filtration respects the approximate boundary between the Kuroshio current and the transition zone. This filtration does not generate many large connected components in the Oyashio current and does not seem to extend the second connected component (in blue), into the current so in that sense it appears to respect that boundary as well. K-means also respects all the expected boundaries, but it also shows extra structure within each region. This hints to one limitation of k-means: the algorithm takes as input the number of desired clusters, which is difficult to know a priori. If the number of clusters is chosen without proper consideration k-means results can be misleading either by showing variation that is not physically meaningful or hiding variation that is. While there are 7 large connected components in the filtration, examination of the local oceans as well as the result of the filtration suggests that three clusters might be more appropriate, one for the Kuroshio current, Oyashio current, and transition zone.





(a) Large connected components with water temperature at 5m depth



(b) Result of a k-means analysis with 4 clusters

Figure 14: Clusters generated via a filtration using the JS distance and via k-means

The fact that these two metrics both find a hole in the same place is not entirely surprising. While it may be a feature of the dataset, both metrics behave very similarly and therefore show similar information. This is conveyed by figure 15, where for each pair of distributions the Jensen-Shannon distance is plotted on the x axis and the Kolmogorov-Smirnov distance is plotted on the y axis. This shows that there is a roughly linear relationship between the two metrics, and any two points that are close to each other in the Kolmogorov-Smirnov sense are likely to also be close to each other in the Jensen-Shannon sense. Since the Kolmogorov-Smirnov distance is entirely determined by the largest difference between the cumulative density functions of a given distribution and the Jensen-Shannon distance is dominated by the largest ratio between the two probability mass functions, and both metrics are bounded, it makes intuitive sense that they would be related in this way.

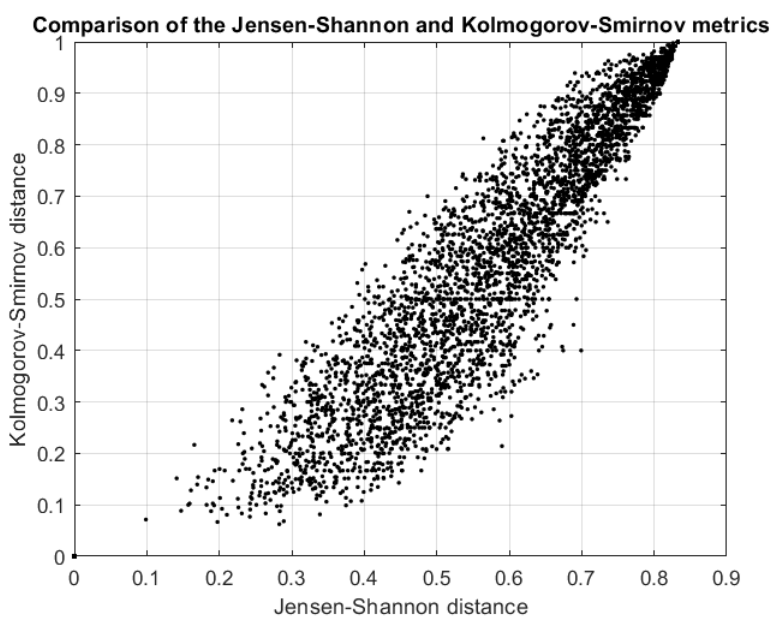


Figure 15: KS distance vs JS distance for each pair of distributions

6.2 Kullback-Leibler Divergence

In our efforts to explain why the results for the Kullback-Leibler Divergence (equation (8)) were well-behaved, we discovered that we had made a significant error in our calculation of the symmetric Kullback-Leibler Divergence. We

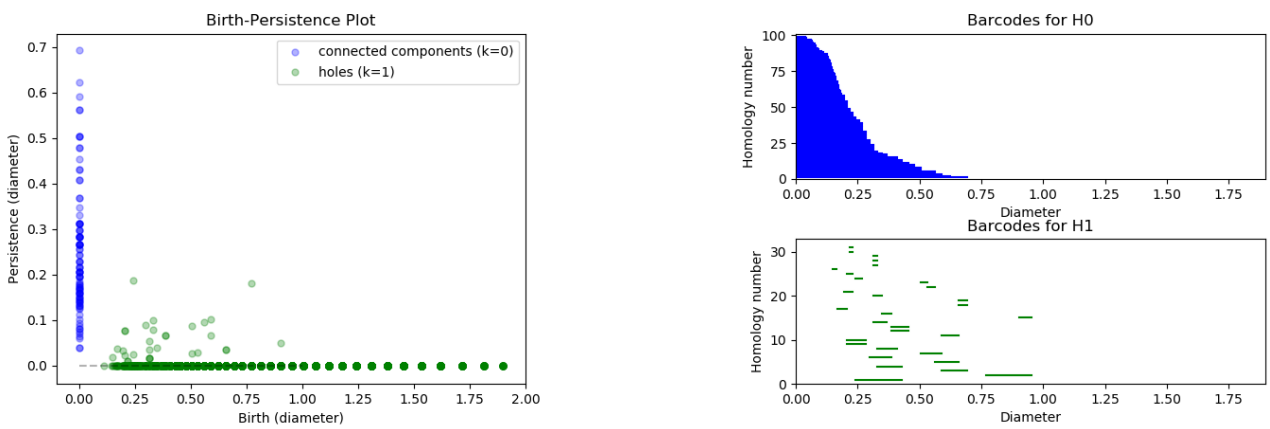
wrote the algorithm to only include finite terms in the sum, which meant that it did not only return a distance of zero if two distributions were identical, but also if they were identical over the intersection of their supports. Consider for example the space $X = \{1, 2, 3, 4\}$ and the two probability mass functions on X given by $f_X(x) = [0, 1/4, 1/4, 1/2]$ and $g_X(x) = [1/2, 1/4, 1/4, 0]$. Since our algorithm would leave all the infinite terms out of the sum, we would be left with the sum over $x \in \{2, 3\}$ and the Kullback-Leibler divergence as we calculated it between f and g would be zero instead of infinite. This is not a property of the Kullback-Leibler divergence itself, only our implementation of it.

Because of this error in our implementation, our version of the symmetric Kullback-Leibler is a premetric rather than a semimetric (which the symmetric Kullback-Leibler divergence over the union of the supports of two distributions is). The proof that our implementation is a premetric rather than a semimetric is as follows:

$D(f_1, f_2) \geq 0$ by our proof of equation (11) since f_1 and f_2 both have non-zero total probability mass over the intersection of their supports. $D(f_1, f_2) = D(f_1, f_2)$ by commutativity of addition. And $(f_1 = f_2) \Rightarrow D(f_1, f_2) = 0$ because if $X^* = f_1^{-1}(Y) \cap f_2^{-1}(Y)$, then $\log \frac{f_2(x)}{f_1(x)} = \log 1 = 0$ for all $x \in X^*$. But $D(f_1, f_2) = 0 \not\Rightarrow (f_1 = f_2)$ in our implementation, because it is not true in general in our implementation that $f_1(x) = f_2(x)$ for all x in the **union** of the supports of f_1 and f_2, since our implementation takes into account only the **intersection** of the supports of the two probability mass functions and leaves out of consideration the points at which either function has zero probability mass.

For this reason, we chose to reprocess the data using the related Jensen-Shannon distance, which also has the advantage of actually being a metric, to replace the section in an earlier draft of our paper that was devoted to the symmetric KL divergence. However, after some discussion we have chosen to briefly discuss the results from our implementation of the KL divergence for completeness since we already discussed them in the presentation. We caution that much more verification would need to be done to present these results beyond the scope of this paper.

First we investigate the persistence diagrams and barcodes derived from doing filtrations on the dataset (figure 16). While, like before, typical 1-homologies have a lifespan much shorter than the longer lived connected components, there are still some 1-homologies showing some signs of persistence. We will focus on the 1-homology appearing at about a diameter of 0.25.



(a) Birth-persistence plot for modified (premetric) KL-divergence

(b) Barcode plot for modified (premetric) KL divergence

Figure 16: Birth-persistence and barcode plots for the filtrations performed using modified (premetric) Kl divergence

A plot of the filtration with diameter 0.31 shows that the hole that we saw previously still appears in the same place as before. This provides more evidence that this hole may be a feature of this dataset rather than a statistical anomaly.

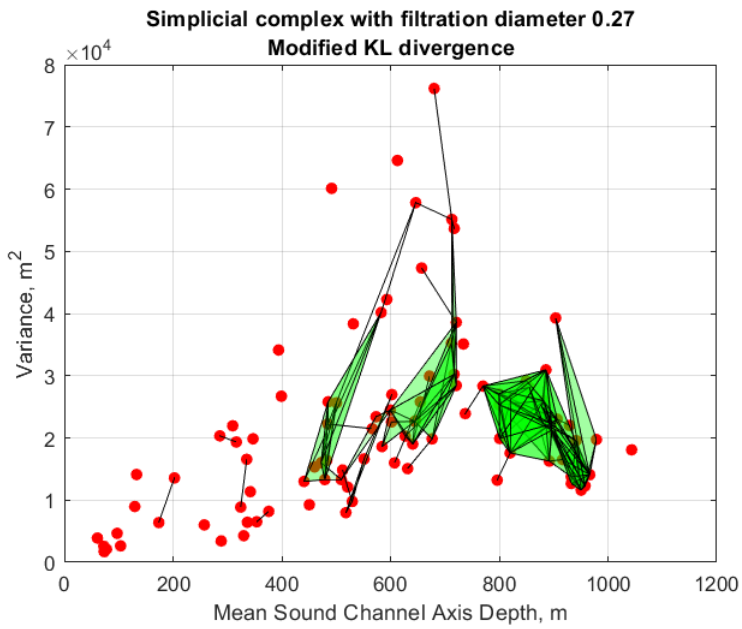


Figure 17: Subcomplex with diameter 0.27 generated using modified KL divergence

Examining these by plotting and color-coding the points on the globe shows the same spatial pattern that we have seen for the other cases. The results seem most similar to the result from k-means with four clusters. The datapoints are split into 4 large groups around 35 deg N (approximately the boundary between the Kuroshio and transition zone) with several smaller outling groups around the boundary between the transition zone and the Oyashio current.

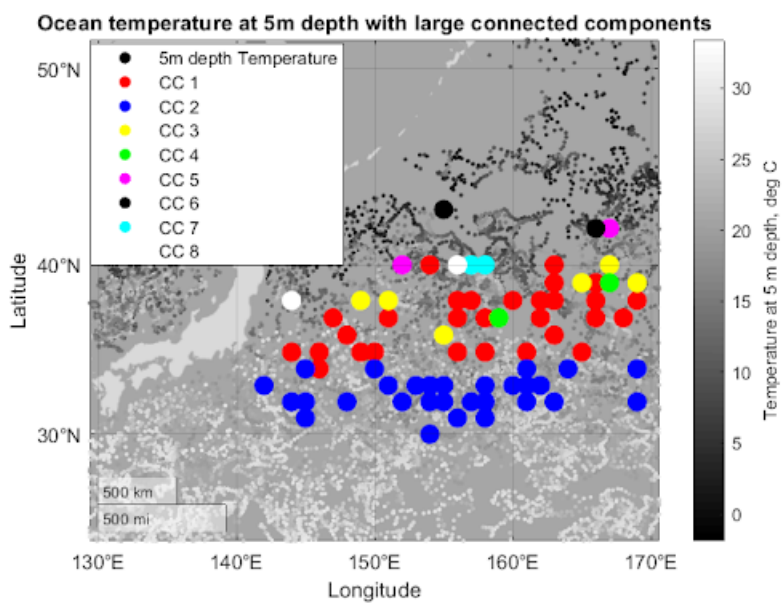


Figure 18: Connected components at diameter 0.26, overlaid on ocean temperature (grayscale) plot

7 Conclusions and Future Work

While we were able to discover somewhat persistent holes in the set of distributions of sound channel axis depth near the coast of Japan, we are uncertain whether this is a feature of the dataset. A physical argument suggests that the large hole could be caused by the currents in the area reducing seasonal variability, since the hole seems to be in the area where there is large variance in sound channel axis depth but only a moderate mean depth, while if there

was seasonal variability we might expect to see data in this hole. Further investigation would be needed, perhaps by looking at similar latitudes that are less affected by strong currents like this area, or by incorporating more data from this region. In addition, an analysis of the raw sound speed profile data might provide hints as to whether this hole actually exists or not.

While our analysis seems to show some response to the intersection of the Kuroshio and Oyashio currents, there are other simpler clustering methods that find this structure as well, such as k-means clustering. In fact, k-means seems to even find more structure than our filtrations. That does not necessarily make persistent homology useless in this situation, for example it could help to inform a decision about the number of k-means clusters to analyze. Since k-means results are sensitive to the number of clusters chosen, this could allow for a more quantitative approach to determining the appropriate number of k-means clusters.

In general, our results support continuation of the methods already used for modeling and classifying oceanographic data. Regular classification algorithms work well, and since there seems to be little underlying structure models do not seem to need to be adjusted to account for any surprising structure in the datasets.

There are several other areas where the techniques of topological data analysis might prove to be fruitful. For example, such techniques might be applied in an attempt to predict when an El Niño might occur from ARGO data by comparing data taken before, during, and after such an event as well as in the years between them. Another potential study might involve looking at data that came from shallow water regions or from following an individual float.

Even restricting ourselves to this dataset there are several other potential opportunities for exploration. For example, it seems common that the biggest connected components consist of two or more distinct clusters that are only connected by a single point. Finding the min-cut (or equivalently, max-flow) of each connected component (as a subgraph) could provide useful insight into this structural property. Several groups are interested in how the paths of the Kuroshio and Oyashio currents change as climate change continues, and these techniques applied over decades might help to track changes in the environment either here or in other areas of the ocean by, for example, comparing persistence diagrams generated on a few years of data at a time. In any case, it seems that there are several potential applications for the techniques of topological data analysis both in the field of underwater acoustics and in oceanography in general.

8 Acknowledgements

The CSEOF reconstructed sea level data was obtained from JPL Physical Oceanography DAAC and developed by the University of Colorado. This data was used to reconstruct sea surface height near Japan from 2007-2009.

9 Appendix

9.1 Proof that D_{JS} is a metric

It is somewhat unintuitive that the Jensen-Shannon distance as given by equation (14) is a metric. To prove that it is, we present the proof from [3].

We denote by $\mathbb{R}^+$ the set of all non-negative real numbers. Now consider the function $L(p,q) : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}^+$ given by

$$L(p,q) = p \log \frac{2p}{p+q} + q \log \frac{2q}{p+q} \tag{15}$$

This is the summand of the Jensen-Shannon divergence. Note that p and q come from a probability distribution, so they are less than or equal to 1. Now say without loss of generality that $p \geq q$. We can rewrite equation (15) as

$$L(p,q) = p \log \frac{p}{p+q} + q \log \frac{q}{p+q} + (p+q) \log 2 \tag{16}$$

Since $p \geq q$, $\frac{p}{2} \leq \frac{p}{p+q} \leq 1$. Now, for fixed p the function $\frac{1}{p+q}$ is a decreasing function of q as can be verified by differentiation. Consider the case where $q = 0$. Then $p \log \frac{p}{p+q} + q \log \frac{q}{p+q} + (p+q) \log 2 = 0 + 0 + \log 2 > 0$. Now say that $q = p$. Then $p \log \frac{p}{p+q} + q \log \frac{q}{p+q} + (p+q) \log 2 = p \log \frac{1}{2} + q \log 12 + (p+q) \log 2 = 0$. So, $L(p,q) \geq 0$ with equality only if $p = q$.

Now recall that Jensen-Shannon divergence is non-negative and equal to zero if and only if $p = q$ and that it is symmetric. The Jensen-Shannon distance inherits these qualities.

Let $p, q, r \in \mathbb{R}^+$. We will now show that $\sqrt{L(p,q)} \leq \sqrt{L(p,r)} + \sqrt{L(r,q)}$. Assume that $p \leq q$ and consider the right hand side $R(r;p,q)$as a function of r. We will show that $R(r)$ has two minima at $r = p$ and $r = q$ and one maximum somewhere between p and q.

First consider the function

$$g(x) = \frac{\log \frac{2}{1+x}}{\sqrt{L(p,q)}} \tag{17}$$

Due to the logarithm in the numerator $g(x)$ changes sign at $x = 1$, and by L'Hospital's rule we can see that $\lim_{x \to 1} g^2(x) = 1$. By differentiation, we see that $g'(x)$ is positive if $f(x) < 0$ with $f(x)$ given by

$$f(x) = \log \frac{2}{1+x} + \log \frac{2x}{1+x} \tag{18}$$

By differentiation, we see that $f(1) = f'(1) = 0$ and $f''(x) < 0$, so we see that $f < 0$ for $x \neq 1$, and $g'(x)$ is positive for $x \neq 1$. Using both of these facts we conclude that $g(x)$ jumps from 1 to -1 at $x = 1$.

Now, let $x = \frac{p}{r}$ and $\beta = \frac{q}{p} \geq 1$. Using this, it can be shown that

$$2\sqrt{r}\frac{\partial R}{\partial r} = g(x) + g(\beta x) \tag{19}$$

Since $|g(x)| \leq 1$ with equality only at $x = 1$ and because $g(x)$ jumps from 1 to -1 at $x = 1$, we conclude that the derivative of $R(r)$ changes sign at $r = p$ and $r = q$. Since $r = \frac{p}{x}$ and $g(x)$ attains a maximum at $x = 1$ we conclude that these are the only two minima, and $\frac{\partial R}{\partial r}$ must attain exactly one maximum between $r = p$ and $r = q$. And since $g'(x) \geq 0$, $g(x) + g(\beta x)$ is monotone increasing for $x \in (\beta^{-1}, 1)$ and has at most one sign change. So, since $r = p$ and $r = q$ are minima of $R(r)$, we have that $\sqrt{L(p,q)} \leq \sqrt{L(p,r)} + \sqrt{L(r,q)}$. Now, by writing equation (14) as

$$D_{JS}(p(x), q(x)) = \sqrt{\frac{1}{2} \sum_{x \in X} \left(\sqrt{L(p(x), q(x)} \right)^2} \tag{20}$$

and applying the Minkowski Inequality, we see that D_{JS} satisfies the triangle inequality. Since it satisfies the other properties of a metric as well, D_{JS} is a metric.

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